

CAROM Frozen-GPT-2 12,000-Update Experiment

Detailed results, failure analysis, and next-run evidence package

CAROM experimental record

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Bottom line. Tripling the run to 12,000 updates did not improve accuracy. Frozen-corpus L2–4 accuracy peaked at 0.327 at step 3,000 and fell to 0.189 by step 12,000, below both the earlier 4,000-step GPT-2 result (0.412) and matched TinyLM result (0.709). Itinerary τ fell from 0.987 to 0.328, coverage from 0.728 to 0.470, and the natural-minus-shuffled endpoint gap from 0.080 to 0.022. Edge prediction remained near 0.70. The collapse begins while the stretched OneCycleLR schedule is approaching its 2×10^{-3} peak, which occurs around step 3,600. This is strong circumstantial evidence for schedule-induced destabilization, not proof: the experiment has one seed and changed schedule trajectory as well as duration.

1 Question, preregistration, and verdict

The experiment asked whether giving the same frozen-GPT-2 CAROM architecture three times as many optimizer updates would raise in-distribution endpoint accuracy while preserving causal trajectory evidence and improving held-out length-5 (L5) execution. The primary target was final L2–4 accuracy above 0.55; the strong target was at least 0.65. Neither target was met. The final accuracy was 0.1888 and the best retained checkpoint reached only 0.3268.

This was not a continuation of the successful 4,000-step checkpoint. It was a fresh seed-0 training run whose OneCycleLR schedule was redefined over 12,000 steps. Consequently, it tests a compound change: more examples *and* a three-times-stretched learning-rate trajectory.

2 Task and mechanism

Each synthetic example contains six symbols in $\mathbb{Z}_8$ and a program of two to five primitive transforms. Natural-language command spans are encoded by the final hidden layer of frozen GPT-2-small (124M parameters, width 768). The trainable CAROM side predicts command dependencies, compiles them into GLV inhibition, routes command representations over reusable neural operators, and executes a 72-step recurrent workspace trajectory. Training uses program lengths 2–4; length 5 is held out as structural extrapolation.

Endpoint slot accuracy, dependency-edge accuracy, repaired itinerary metrics, causal trajectory replacements, and an L5 integration-budget sweep are all reported. High itinerary τ is never read alone: an incomplete trajectory can preserve pairwise order among the few modes it visits.

3 Exact protocol and execution history

Base encoder	Frozen GPT-2-small, final hidden layer, width 768
Seed / batch	Seed 0; batch 64; 768,000 generated training examples
Optimizer	AdamW; max LR 0.002; weight decay 10^{-4} ; gradient clip 1.0
Schedule	PyTorch OneCycleLR over 12,000 steps; default 30% warm-up
Checkpoints	Step 0 and every 1,000 steps through 12,000
Frozen evaluation	L2-4: seed 1234, $n = 256$; L5: seed 5678, $n = 128$
Interventions	Natural, forced-correct, shuffled, and smeared trajectories
L5 budgets	72, 100, and 120 recurrent integration steps
Hardware	RunPod Secure Cloud; one A100 SXM4 80GB; USD 1.49/hour
Training time	10,582 seconds (2h56m22s)
Final pod	ws1lxvsshf7j1; terminated after verified retrieval

Two earlier launch attempts were automatically deleted because stale “awaiting approval” prose remained in experiment records. The third launch completed. Those failures consumed money but did not alter the final run’s weights. The automatic post-training evaluator then inherited an older corpus unpacking bug (expected seven fields, received five). A previously tested repair was uploaded and an intervention-only continuation evaluated all 13 checkpoints without retraining. The continuation exited zero.

The final archive contains 13 checkpoints, 13 result records, source and logs; all 31 files pass the retrieved SHA-256 manifest. Total archive size is about 152 MiB.

4 Learning-rate trajectory and online training telemetry

OneCycleLR’s meaning changes when total steps change. In the 4k run the maximum LR occurred near step 1,200 and had fallen to 5.65×10^{-4} by step 3,000. In the 12k run, LR was still rising at step 3,000 and peaked near step 3,600.

Table 1: Training telemetry and reconstructed 12k OneCycleLR. Online metrics use the training process’s stochastic evaluation stream; frozen results follow.

Step	LR	Loss	Online acc	Online τ
1000	.000423	2.177	.222	.997
2000	.001207	1.931	.302	.988
3000	.001872	1.781	.312	.953
4000	.001989	2.394	.190	.816
5000	.001866	3.192	.126	.728
6000	.001623	2.203	.182	.320
7000	.001294	2.159	.195	.263
8000	.000925	2.142	.199	.242
9000	.000566	2.193	.201	.305
10000	.000267	2.228	.174	.297
11000	.000069	2.152	.204	.250
12000	8×10^{-9}	2.176	.188	.242

Observed. Loss is lowest and accuracy highest at step 3,000. During the next 2,000 steps, near maximum LR, loss rises to 3.192 and accuracy drops to 0.126. Later annealing does not restore the earlier solution.

5 Frozen checkpoint panel

Table 2: Fixed-corpus results. Acc = L2–4 endpoint accuracy; gap = natural minus shuffled endpoint accuracy.

Step	Acc	Edge	τ	Cov.	Exact	Trans. rec.	Gap	L5
0	.139	.297	.040	.383	.000	.017	.000	.135
1000	.224	.703	.997	.715	.313	.429	.001	.169
2000	.292	.703	.996	.714	.316	.443	.026	.219
3000	.327	.703	.987	.728	.332	.448	.080	.262
4000	.167	.703	.793	.634	.231	.348	.004	.151
5000	.126	.713	.814	.644	.238	.351	-.011	.121
6000	.182	.716	.413	.509	.141	.183	.009	.156
7000	.184	.708	.352	.480	.102	.141	.011	.143
8000	.188	.720	.336	.474	.102	.136	.022	.158
9000	.191	.714	.332	.472	.098	.132	.035	.160
10000	.195	.725	.328	.470	.094	.128	.032	.152
11000	.186	.726	.332	.472	.098	.132	.025	.162
12000	.189	.731	.328	.470	.094	.128	.022	.159

Observed. The best checkpoint is step 3,000 on endpoint accuracy, trajectory coverage, exact order, transition recall, causal gap, and L5 accuracy. Edge accuracy is almost unchanged through the initial collapse and actually rises to 0.731 at the final checkpoint. Thus dependency prediction survived while the downstream routing/execution regime degraded.

6 Trajectory interventions

Table 3: L2–4 endpoint accuracy under trajectory replacements.

Step	Natural	Forced	Shuffled	Smeared	Nat–shuf
0	.139	.139	.139	.139	.000
1000	.224	.222	.223	.222	.001
2000	.292	.265	.266	.264	.026
3000	.327	.277	.247	.263	.080
4000	.167	.167	.163	.165	.004
5000	.126	.136	.137	.137	-.011
6000	.182	.173	.173	.173	.009
7000	.184	.173	.173	.173	.011
8000	.188	.166	.166	.166	.022
9000	.191	.156	.156	.156	.035
10000	.195	.163	.163	.163	.032
11000	.186	.161	.161	.161	.025
12000	.189	.167	.167	.167	.022

At step 3,000 the natural trajectory is materially useful: shuffling reduces accuracy by 0.080. This evidence nearly disappears at step 4,000, becomes slightly negative at step 5,000, and only weakly recovers. Forced-correct is not an oracle because it also changes dwell and amplitude statistics. These results show causal trajectory dependence, not learned commutation; commutation requires the separately controlled nonlinear swap and JVP probes.

7 Held-out L5 and integration budget

Table 4: L5 endpoint accuracy by recurrent integration budget.

Step	S=72	S=100	S=120
0	.135	.135	.135
1000	.169	.171	.169
2000	.219	.227	.220
3000	.262	.262	.253
4000	.151	.142	.135
5000	.121	.117	.124
6000	.156	.147	.150
7000	.143	.146	.141
8000	.158	.160	.158
9000	.160	.163	.161
10000	.152	.159	.154
11000	.162	.169	.160
12000	.159	.171	.167

Longer integration does not rescue the model. At the best checkpoint, S=120 is slightly worse than S=72. The failure is therefore not well explained by insufficient inference settling alone.

8 Comparison with prior arms

Model/checkpoint	L2-4	τ	Coverage	Exact	Nat-shuf	L5
GPT-2 12k, best (3k)	.327	.987	.728	.332	.080	.262
GPT-2 12k, final	.189	.328	.470	.094	.022	.159
GPT-2 4k, final	.412	.904	.710	.297	.115	.214
TinyLM 4k, final	.709	.825	.911	–	.227	.356

The earlier GPT-2 4k run is better on L2-4 accuracy and causal gap, while the 12k step-3,000 checkpoint is modestly better on L5 and itinerary completeness. Neither GPT-2 arm approaches TinyLM endpoint performance.

9 Failure analysis

Observed. Three signatures coincide: endpoint loss worsens immediately after step 3,000; itinerary structure collapses; and the natural-trajectory advantage vanishes. Edge accuracy does not collapse. This localizes the damage downstream of, or orthogonal to, edge supervision.

Inference. The most economical explanation is that stretching OneCycleLR exposed the model to a long high-LR interval and displaced the useful step-3,000 basin. At step 3,000 the 12k run uses LR 0.001872, versus 0.000565 at the same step of the 4k schedule. At step 4,000 the 12k run is near maximum LR, whereas the 4k run has annealed to effectively zero. The timing is unusually specific.

Hypothesis. Other live explanations are (i) a seed-0 optimization bifurcation not reproducible across seeds; (ii) high LR interacting particularly with routing or GLV parameters; (iii) loss-term imbalance, since edge performance stays high while endpoint behavior fails; and (iv) a training/evaluation mismatch. The present single run cannot apportion these effects.

10 What the next experiment must distinguish

This report intentionally does not declare one replacement configuration. The independent reviewers are asked to rank designs that separate duration from LR schedule, preferably with inexpensive gates:

1. resume the successful 4k checkpoint with a low constant or cosine-decay LR, preserving the known basin;
2. repeat a fresh 12k run with lower peak LR and early stopping on the fixed corpus;
3. use parameter-group LR or freeze the already-solved edge head;
4. replicate the leading setting across seeds before mechanistic claims;
5. add gradient norms, per-loss terms, routing entropy, and repaired swap/JVP probes so the next failure is identifiable while it happens.

Any next run should checkpoint frequently through steps 2,000–5,000, monitor a fixed validation panel, and stop when both endpoint accuracy and causal gap degrade persistently. Selection must be declared in advance to avoid treating the best of many checkpoints as an unbiased final result.

11 Limitations and epistemic status

- One seed gives no variance estimate.
- Training duration and LR trajectory changed together.
- The online and frozen evaluations use different sample streams; only the frozen panel is used for checkpoint comparisons.
- The forced intervention also changes dwell/amplitude, so it isolates more than order.
- No repaired v3 commutator, schedule-stability, or decisiveness probe was run; no claim of learned commutation is licensed.
- TinyLM comparison values are from the matched earlier record, not rerun in this experiment.

12 Reproducibility and evidence

The authoritative run record is `projects/carom/experiments/20260722T015200Z-carom-gpt2-12k/`. It contains the exact runner, launch command, training and intervention logs, all 13 checkpoints, per-checkpoint JSON, summary JSON, and `artifacts.sha256`. Verification confirmed the step set $\{0, 1000, \dots, 12000\}$, required result schema, checkpoint count, and every manifest digest before remote termination.